

Multiple Choice

1. **Answer:** A

Options: A) convolutional networks; B) graph networks; C) fully connected networks; D) RBF networks

Note: Correct option: A - convolutional networks

2. **Answer:** B

Options: A) they are biased; B) they have high variance; C) they are not consistent estimators; D) None of the above

Note: Correct option: B - they have high variance

3. **Answer:** A

Options: A) linear in K; B) quadratic in K; C) cubic in K; D) exponential in K

Note: Correct option: A - linear in K

4. **Answer:** D

Options: A) To save computing time during testing; B) To save space for storing the Decision Tree; C) To make the training set error smaller; D) To avoid overfitting the training set

Note: Correct option: D - To avoid overfitting the training set

5. **Answer:** A

Options: A) PCA; B) Decision Tree; C) Linear Regression; D) Naive Bayesian

Note: Correct option: A - PCA

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The use of prior distributions on the parameters in a probabilistic model.'

Long Form Response

11. **Answer:** `def nth_items(list,n): | return list[:n]`

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def divisible_by_digits(startnum, endnum): | return [n for n in range(startnum, endnum+1)]`

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key